



INTERNSHIP PROGRAM 2023

PROJECT REPORT

MACHINE LEARNING

GRADUATE CAREER RECOMMENDER SYSTEM

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- The **text between inequality (< >)** is to be replaced by relevant text
- Please **remove the yellow highlight on the Text** between the inequality (< >). This is done to help you notice the text to be changed/replaced
- The text in *italics* highlighted in grey is just for reference and should be removed after adding the relevant text

1 PROJECT DETAILS

Project Name	Graduate Career Recommender System		
Project Sponsor	Tushar Topale		
Project Manager	Harshada Topale		
Start Date	07-03-2025	Completion Date	20-06-2025

2 SUMMARY

Graduate Career Recommender is a web portal primarily for students, faculty, and admins, which aims to suggest appropriate & suitable job roles to user depending on their inputs. Like aptitude scores, hobbies & interests, and career/academic preferences. The project was undertaken to provide a concise and one stop solution for suggesting most appropriate & suitable job role to undergrad students in their final year. Eliminating the dilemma and confusion between picking their desired job role, which is often fuelled & influenced by peer pressure. In long term this portal will keep delivering clarity in decision making to our future engineers, doctors, architects, artists, etc., regarding possible & suitable career.

3 INTRODUCTION

3.1 Background

In the ever-growing world of science & technology, and with the advent and its fast-paced adoption in almost every industry, we will see old mundane jobs perish and other existing & new jobs roles leveraging AI emerge. In such a dynamically changing job market, undergraduate students need to set their career priority first hand. It would be even better if they set their career choice before getting into university/college. However, students in their final year of college need to hurry up finally choosing their desired job role.

Still, we get to see cases where students are unable or regret their desired job role choice later, just because their friend/peer took or ask them to take that career. Such crucial career/life decisions must not be taken under anybody's influence but should be solely the student's personal unbiased preference/choice.

Students must choose a career only after assessing their strengths, weaknesses, aptitude, and their personal preferences or likings of what they like to do or work on. Well, this can only be known after trying working on some small DIY projects or prototypes. In the process of building these prototypes students will learn what they enjoy working on, what are their weak areas, and where they exceed their expectations of work.

In this exhaustive journey of finding the dream job role, we have found a way to help students identify their desired career using Machine Learning. This help may not be with building prototypes/projects but with making the final choice. We came up with a job role recommendation portal, which suggests

the most appropriate job role to the student after assessing his/her job preferences, aptitude, and other parameters. The parameters are same on which the machine learning models were trained. Based on the models suggested job roles the portal recommends some free online courses, from popular websites like edx, Coursera, YouTube, etc. Students can then refer these courses to begin their preparation towards securing their desired job role.

3.2 Stakeholders

Undergraduate Students: Students will be greatly benefited by this initiative as they will clear and concise job role suggestion & course recommendation after analysing their strength, weakness, aptitude, and career preferences.

Recruiters/Institutions: Recruiters & education institutions can leverage this portal as a free tool for their students to help them identify their desired career/job role.

3.3 Objectives

The solution involves developing a predictive model and a web application for your graduation students, which will suggest an appropriate job role and a course which help them secure the suggested job role. The model will utilize essential historical data for predicting an appropriate job role for the student, and the app will suggest the required course. The model will be trained on dataset containing students' academic records and their career/subject preferences.

4 METHODOLOGY

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4.1 Considerations & Assumption

The provided dataset has relevant features for target variable prediction. Students' information integrity. Students self-reported skills, rating, and preferences are assumed to be true and genuine. Library compatibility. All libraries to be used are assumed to be compatible with each other and available in the environment. Artifacts persistence. Saved artifacts, encoders, models, and metrics are available for future references and deployment purposes.

4.2 Approach

Here the approach we followed to solve the problem & finally build the portal:

4.2.1 Data Preprocessing:

The primary, most essential, and time-consuming step in training any ML model is data preprocessing. No model can be trained without preprocessing data on which the model is to be trained. Mentioned below are the preprocessing & feature engineering techniques which were used:

- Analysing dataset shape, features & target variables and their data types, finding number of null values present in the dataset.
- Performing [Exploratory Data Analysis](#) on the data set to understand the relationship amongst the feature variables and target.
- Used [Label Encoding](#) & [One Hot Encoding](#) to encode categorical features because ML models are unable to process string or character data.
- Used [Feature Engineering](#) to creating new [Interaction Features](#). [Interaction Features](#) are combination of existing features in the dataset. Sometimes, these new features, which were ideally not present in the dataset, help model learn, predict, generalize better on new unseen data.
- [Feature Selection](#). Identifying most important features using [SelectKBest](#) & chi-square ([chi2](#)) scores.
- [Under-sampling](#) dataset. As the dataset provided was unbalanced, means the number of target classes samples which were to be predicted by the ML model were unequal. In such cases the trained model gives biased predictions. Because it learns the majority class very well as they are available in large numbers and performs poorly against the minority classes.

4.2.2 Model Selection & Training:

- We choose to train two ML models: [Random Forest Classifier](#) & [Neural Network](#) model.

Why Random Forest (RF): *It's inherently hard to overfit out of the box. Random Forest has a "self-regularization" mechanism which restricts it from overfitting the data. As RF is an ensemble of Decision Trees, each tree receives a sub-set of training data which reduces the chances of each tree to overfit the data. Also, as the number trees increases the model variance decreases without increasing the model bias. And it does not require feature scaling, which makes the model solution less error prone, including production pipeline. So, these factors make Random Forest a go-to baseline model.*

Why Neural Network (NN): *Neural Network are good choice for classification task when dealing with high-dimension data where traditional methods struggle to find patterns. They can find patterns in data that may not be obvious to humans or traditional algorithms. With sufficient data and appropriate architecture NN can out-perform traditional ML models. NN offer better scalability, hence good choice for production pipelines.*

- Splitting data in to training, validation, and test set. With following split ratio> `train:validation:test = 0.8:0.1:0.1`
- We used [Optuna](#), [StratifiedKFold](#), [make_pipeline](#) python libraries for training the RF model and [PyTorch](#) for the NN model.

- [MLFlow](#) used for experiment tracking, metrics & artifacts logging, and for model deployment.

4.2.3 Model Evaluation:

Following metrics used for evaluating RF and NN model performance:

[Accuracy](#): Percentage of overall correct prediction in the respective dataset.

[Balanced Accuracy Score](#): Average of recall & specificity. Better metric for imbalance classes case, where difference between class samples is greater.

[Precision](#): Percentage of predicted positives are correct. Critical when false positives as costly. When the model predicts an incorrect job role.

[Recall](#): Percentage of actual positives correctly identified. Important when false negatives are risky. When the model misses an appropriate job role.

[F1 Score](#): Harmonic mean of precision & recall. Best for imbalance data where both errors matter, namely precision and recall.

[ROC-AUC](#): Measures the separation of positive and negative classes. Primarily for binary classification but can be adjusted for multiclass classification.

[PR-AUC](#): Focuses on performance of positive class. Better than ROC-AUC for higher class imbalance, where difference between class samples is greater.

[Matthews Correlation Coefficient \(MCC\)](#): It measures the correlation between predicted and actual values, considering all 4 elements of confusion matrix. MCC provides a balanced measure, even for imbalanced classes, and ranges from -1 to $+1$, where $+1$ indicates perfect prediction, 0 represents random prediction, and -1 indicates a complete disagreement between predictions and actual labels.

4.2.4 Finalizing Model for Deployment:

After training and experimenting with numerous hyper-parameters 3 best performing [Random Forest Classifier](#) and 1 [Neural Network](#) model were chosen for deployment using [Streamlit](#) Cloud. The selected models were logged to [MLFlow/DAGsHub](#) repository.

4.2.5 Model Deployment:

A fully featured web app was designed using [Streamlit](#) library. An intuitive design suitable for both students and admins was adopted. The selected logged models were installed in the web app and deployed on [Streamlit](#) Cloud.

4.3 Activities

To deliver the project above mentioned approach was adopted including the following key activities:

4.3.1 Requirements Gathering:

Conducted group discussions with key stakeholders, including students, faculties, career coach/counsellors, and industry experts to identify project scope, sources to gather data, critical features needed for training ML model.

4.3.2 Planning:

An exhaustive project plan was developed with detailed timeline & milestones to be achieved under product development. The plan also allocates essential resources required for the project and outlines potential risks.

4.3.3 Data Collection and Cleaning:

The available dataset, including students' interests, aptitude, subject and career preferences, was thoroughly processed handling missing values and data type inconsistencies.

4.3.4 Data Pre-processing & Feature Engineering:

Categorical features were numerically encoded to be processed by ML models. New interaction features were created boosting models' performance by 30%. Important and relevant features for model training were identified using [SelectKfold](#) and [chi2](#) techniques. Under-sampled the dataset to balance the target classes.

4.3.5 Model Training:

Trained Random Forest Classifier and Neural Network model. Used [Optuna](#), combined with [StratifiedKFold](#), for finding optimal hyper-parameters for RF model, as it's faster than [GridSearchCV](#). [PyTorch](#) was used for building & training NN model. [MLFlow](#) helped in experiment, metric, and artifact logging.

4.3.6 Web App Integration:

The web app portal was built with an intuitive design in mind for students, faculties, and admins. The finalized models were successfully integrated in the web app. The final web app portal was deployed on [Streamlit](#) Cloud for testing and deployment purpose.

4.3.7 Testing:

The deployed web app was thoroughly tested for all edge cases and was deployed to production after confirmation from Team Lead.

5 TARGETTED V/S ACHIEVED OUTPUT

Targeted Output: The ultimate objective was to build a working solution/web app with integrated ML model which will suggest an appropriate job role to user after processing the details feed by them. Depending upon the suggested job role the web app will recommend some suitable free course links from popular websites like Coursera, edX, YouTube, etc. to help user get started with their preparations.

The ML models are expected to perform with high accuracy, precision, and recall after training on dataset provided.

Achieved Output: The web app performance was satisfactory. But both the model's performance was poor, as expected after considering their validation & test set metrics. On thorough examination and continuous discussions, it was found that due to limited/inadequate data and less impactful features both models were unable to fully learn the dataset. Leading to poor and random predictions on validation and test set, and ultimately on unseen live data.

6 CONCLUSION

Considering the current performance of models, they are deployable, but their predictions or job role suggestions cannot be blindly trusted. As for our problem statement we used a balanced approach using both precision & recall and knowing models' poor recall. Where the chances of the models on missing a suitable job role for a candidate are quite high, along with wrong job role suggestions. Completely trusting such models is not recommended. But they are workable. Our recommendation to the stakeholders is that users can use the portal, but users' discretion is advised.

On special note, Users may pay special attention to the models' confidence score after getting their job role suggestions. If the confidence score lies in a range of 37%-39%, Users may consider it a correct prediction/suggestion. But must also review the parameters feed to the models, if they justify the predictions/suggestions. Users may still seek advise from experts regarding models' suggestions. As in any case genuine human feedback is always invaluable.

The scope of this project can be expanded to any field of study. Currently, it only deals with engineering and technology disciplines, but it can be extended to different domains like management, hospitality, medical, arts, entertainment industry, etc., there are endless possibilities. The only roadblock is *relevant superior quality data having relevant features in good quantity of respective domain.*

7 APPENDICES

7.1 Appendix A – Project Deliverable List

Sr. No.	Deliverable Name	Status
1	Development Log	Completed
2	GitHub Repository/Source Code	Completed
3	Lessons Learned Log	Completed
4	Project Report	Completed
5	Project Schedule	Completed
6	RAID Log	Completed
7	Project Video	Completed
8	Work Breakdown Structure	Completed